

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)			The vibrations / oscillations (1) are parallel to the direction of wave or travel or energy transfer (1)	2			2		
	(b)			Speeds of 9 OR 5 [km/s] OR 0.5×8 squares (1) Difference = 4 [km/s] (1)		2		2	1	
	(c)	(i)		[Maximum] speed in the mantle is greater than the speed in the outer core / greatest speed is in the mantle / 15 [km/s] and 13 [km/s] (1) Mantle acts like a solid / outer core is liquid (1) so Bob's claim is true. Conclusion must be included to award 2 marks			2	2		
		(ii)		Mean speed = $\frac{6\,300}{550}$ (1) = 11.45 (km/s) (1) This is the actual speed at depth of 1 200 km (or at 4 900 km) (1) [So Bob's statement is not true.] Alternative for third mark: [At 3 500 m] the speed is 10.0 [km/s] [So Bob's statement is not true.] Alternative: Speed at 3500 km is 10.0 [km/s] (1) Time = $\frac{6\,300}{10}$ (1) = 630 [s] (1) [So Bob's statement is not true.] Alternative: Speed at 3500 km is 10.0 [km/s] (1) Distance = 10×550 (1) = 5 500 [km] (1) [So Bob's statement is not true.]			3	3	2	

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					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		$2 \times 550 = 1\,100$ [s]		1		1	1	
		(ii)		P wave only shown i.e. one cycle (1) Size no bigger than wave at B (1) Position, the start of the wave must be within the correct 200 s range based on (d)(i) (expect 1 000 – 1 200) or apply an ecf (1) N.B. If drawn correctly but at station A or B apply a 1 mark penalty			3	3		
				Question 8 total	2	3	8	13	4	0